

LECTURE-15

Wednesday (online)
25/3/20.

JOINT PROB. DISTRIBUTION:

⇒ Univariate Distri. Func.: Single random variable distribution. $f(x)$

⇒ Bivariate ^{Prob.} Distri Func.: Two random variable distri. func. $F(x, y)$

$$f(x_i, y_j) = P(X = x_i * Y = y_j) \quad \begin{matrix} \text{random variables} \\ \text{values} \end{matrix} \quad \begin{matrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \end{matrix}$$

X \ Y	y ₁	y ₂	y ₃		y _m
x ₁	$f(x_1, y_1)$	$f(x_1, y_2)$	$f(x_1, y_3)$		$f(x_1, y_m)$
x ₂	$f(x_2, y_1)$	$f(x_2, y_2)$	$f(x_2, y_3)$		$f(x_2, y_m)$
⋮	⋮	⋮	⋮	⋮	⋮
x _m	$f(x_m, y_1)$	$f(x_m, y_2)$	$f(x_m, y_3)$		$f(x_m, y_m)$
P(Y=y _j)	$h(y_1)$	$h(y_2)$	$h(y_3)$		$h(y_m)$

X \ Y	y ₁	y ₂		y _m	P(X=x _i)
x ₁	$f(x_1, y_1)$	$f(x_1, y_2)$		$f(x_1, y_m)$	$g(x_1)$
x ₂	$f(x_2, y_1)$	$f(x_2, y_2)$		$f(x_2, y_m)$	$g(x_2)$
⋮	⋮	⋮	⋮	⋮	⋮
x _m	$f(x_m, y_1)$	$f(x_m, y_2)$		$f(x_m, y_m)$	$g(x_m)$
P(Y=y _j)	$h(y_1)$	$h(y_2)$		$h(y_m)$	<u>1</u>

$P(X=x_i)$ and $P(Y=y_j)$ are marginal Probabilities (Univariable)
 marginal prob. of x marginal prob. of y.

⇒ Joint Prob. Function should hold these conditions.

① $f(x_i, y_j) > 0$

② $\sum_i \sum_j f(x_i, y_j) = 1$

⇒ Finding marginal prob. of Y from joint Prob. distr. func. $f(x_i, y_j)$

$$h(y_j) = \sum_{i=1}^m f(x_i, y_j) \Rightarrow P(Y=y_j)$$

jo nahi chia
hai (i.e. n)

$$= f(x_1, y_j) + f(x_2, y_j) + \dots + f(x_m, y_j)$$

wo idhr change hoga like below

[eg.]

$$x = 0, 1, 2$$

$$h(3) = f(0, 3) + f(1, 3) + f(2, 3)$$

$$= 1/13 + 3/13 + 8/13 = \underline{\text{Ans.}}$$

⇒ Conditional Prob. Func.

→ X given Y

$$f(n_i | y_j) = P(n_i | y_j) \\ = \frac{P(X=n_i \text{ and } Y=y_j)}{P(Y=y_j)} \\ = \frac{f(n_i, y_j)}{h(y_j)}$$

⇒ Independent bivariate variables: IF joint prob func = Product of its marginals.
 $f(n, y) = g(n) h(y)$

Q: A bag contain 3 black, 2 red, 3 green balls. 2 balls are selected at random. If X is no. of black balls, Y is no. of red balls then find Total = 8 balls

(a) The joint probability function: $f(n, y)$.

$$f(n, y) = \frac{{}^3C_n {}^2C_y {}^3C_{2-n-y}}{{}^8C_2}$$

→ two chosen so max.
 $n=0,1,2$
 $y=0,1,2$
 $0 \leq n+y \leq 2$

after putting values of n and y in above joint func.

X \ Y	0	1	2	$P(X=x_i) \rightarrow g(n)$	Sum of row
0	3/28	6/28	1/28	10/28	
1	9/28	6/28	0	15/28	
2	3/28	0	0	3/28	
$P, Y=y_j \rightarrow h(y)$	15/28	12/28	1/28	1	

OR

(n, y)	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(2,0)
$f(n, y)$	3/28	6/28	9/28	1/28	1/28	3/28

Marginal func: for $g(\frac{\text{value of } n}{})$, fix n and find y
 for $h(\frac{\text{value of } y}{})$, fix y and find n.

(b) $P(X+Y \leq 1)$. Find this prob.
 * From Prev. table

$$P(X+Y \leq 1) = P(0,0) + P(0,1) + P(1,0) \\ = \frac{3}{28} + \frac{6}{28} + \frac{9}{28} \\ = \frac{18}{28}$$

(c) Marginal prob. distr. of $g(n)$ and $h(y)$:

n	0	1	2
$g(n)$	10/28	15/28	3/28

y	0	1	2	last row of table
$h(y)$	15/28	12/28	1/28	

from last column of joint pd table
 $g(0) = f(0,0) + f(0,1) + f(0,2) = 10/28$

$$h(0) = f(0,0) + f(1,0) + f(2,0) = 10/28$$

(d) Find Conditional p.d $f(n | 1)$

$$f(n | 1) = \frac{f(n, 1)}{h(1)}$$

formula on prev. page in pink

$$h(1) = \frac{6}{28} + \frac{6}{28} + 0 \quad \text{from middle column (where } y=1) \\ = \frac{12}{28} = \frac{3}{7}$$

$$f(n | 1) = \frac{f(n, 1)}{3/7}$$

$n=0,1,2$.

→ value of $f(0,1)$ from joint pd table.

$$f(0,1) = \frac{7}{3} \left(\frac{6}{28} \right) = \frac{1}{2}$$

$$f(1,1) = \frac{7}{3} \left(\frac{6}{28} \right) = \frac{1}{2}$$

$$f(2,1) = \frac{7}{3} (0) = 0$$

answer:

n	0	1	2
$f(n 1)$	1/2	1/2	0

(c) $P(X=0 | Y=1)$ and are X & Y independent?

* From answer table of part (d)

$$P(0|1) = \boxed{1/2}$$

* For independence; if $f(u,y) = h(y) \cdot g(u)$ then ✓

• Checking at $(0,0)$. (we can check any cell)

$$g(u) \text{ at } (0,0) = 10/28$$

$$h(y) \text{ at } (0,0) = 15/28$$

$$f(u,y) \text{ at } (0,0) = 3/28$$

$$g(u) \cdot h(y) = 10/28 \times 15/28 = 75/392$$

* Since $75/392 \neq 3/28$ $\therefore$ They are ^{NOT} independent.

Q: Joint p.d fnc. for two discrete random variable X & Y :

$$f(u,y) = \frac{xy^2}{30} \quad u=1,2,3 \text{ & } y=1,2$$

Are X & Y independent?

Marginal P.D for X :

$$g(u) = \sum_{y=1}^2 \left(\frac{xy^2}{30} \right)$$

$$g(u) = \frac{u(1)^2}{30} + \frac{u(2)^2}{30} = \boxed{\frac{u}{6}}$$

Marginal P.D for Y :

$$h(y) = \sum_{u=1}^3 \left(\frac{xy^2}{30} \right)$$

$$h(y) = \frac{1y^2}{30} + \frac{2y^2}{30} + \frac{3y^2}{30} = \boxed{\frac{y^2}{5}}$$

* If $f(u,y) = h(y) \cdot g(u)$ then independent $\therefore$

$$\frac{xy^2}{30} = \left(\frac{x}{6} \right) \cdot \left(\frac{y^2}{5} \right)$$

$$\frac{xy^2}{30} = \frac{xy^2}{30}$$

$\therefore X$ & Y are independent ✓

Q:

$y \backslash u$	2	4	$h(y)$
1	0.10	0.15	0.25
3	0.20	0.30	0.50
5	0.10	0.15	0.25
$g(u)$	0.40	0.60	1.00

Find $E(X), E(Y), E(X+Y), E(2X-3Y)$ and $E(XY)$

→ made for solution

$$E(X) = \sum (u_i \cdot g(u_i)) = [u_2 \cdot g(u_2)] + [u_4 \cdot g(u_4)] \rightarrow (2 \times 0.40) + (4 \times 0.60) = 3.2$$

$$E(Y) = \sum (y_j \cdot h(y_j)) = (1 \times 0.25) + (3 \times 0.50) + (5 \times 0.25) = \boxed{3.0}$$

$$E(X+Y) = \sum_i \sum_j (u_i y_j) \cdot f(u_i, y_j)$$

collision cell of $y=2, u=1$

$$= (1+2) \times (0.10) + (2+3) \times 0.20 + (2+5) \times (0.10) + (1+4) \times (0.15) + (3+4) \times 0.30 + (4+5) \times (0.15) = \boxed{6.20}$$

Note: $E(X+Y) = E(X) + E(Y)$
 $6.20 = 3.0 + 3.2$ ✓

$$\begin{aligned}
 E(2x - 3y) &= 2E(x) - 3E(y) \\
 &= 2(3.2) - 3(3.0) \\
 &= \boxed{-2.6}
 \end{aligned}$$

CO-variance of Two Random variables: It is a measure to tell how two variables changes together.

$$\text{Cov}(X, Y) > 0$$

Both increase together

$$\text{Cov}(X, Y) < 0$$

Both decrease together

$$\text{Cov}(X, Y) = 0$$

Both are Independent.

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$\Rightarrow$ Correlation Coefficient: A better form of covariance that tells how strongly the two variables are related (not just increase/decrease).

$$\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

$\rightarrow$ It is b/w -1 & $+1$ where: $+0.9 =$ Strong increase
 $-0.8 =$ negative strong decrease.